Two-Column Academic Paper: A Study in Reading Order Preservation

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Abstract

This paper presents a two-column layout fixture designed to test whether PDF parsing tools correctly preserve reading order. The left column should be read before the right column on each page. The introduction continues in the left column of the next section.

1. Introduction

Knowledge-intensive NLP tasks such as question answering, fact checking, and dialogue require access to external information. Retrieval-augmented generation (RAG) systems address this by grounding generation in retrieved documents.

The study of information retrieval has a long and rich history in computer science. Early systems relied on boolean queries and inverted indices to match documents to user needs. With the advent of statistical methods and machine learning, ranking functions became more sophisticated.

In recent years, neural retrieval models have achieved remarkable improvements. Dense passage retrieval using bi-encoder architectures enables semantic matching beyond lexical overlap. Cross-encoder rerankers further improve precision at the cost of additional computation.

However, naive RAG systems struggle with complex queries that require reasoning across multiple documents. Graph-based approaches have shown promise in aggregating evidence from heterogeneous sources.

The transformer architecture has become the dominant model for both retrieval and generation tasks. Pre-trained language models such as BERT and T5 provide strong initialization for domain-specific fine-tuning. Scaling laws suggest that performance continues to improve with model size and training data.

Despite these advances, challenges remain in handling long documents, multi-modal content, and cross-lingual retrieval. The cost of deploying large models at inference time is also a significant practical concern. Efficient architectures such as ColBERT and PLAID aim to address these issues.

2. Related Work

Neural retrieval models have shown impressive performance gains over traditional methods. DPR, ANCE, and ColBERT represent different points in the design space of dense retrieval systems.

Graph neural networks have been applied to various NLP tasks including relation extraction, document classification, and multi-hop reasoning. Recent work explores combining retrieval with graph-based reasoning.

3. Experimental Results

We evaluated our approach on three standard benchmarks: MultiHopQA, HotpotQA, and MuSiQue. Table 1 summarizes the main results. Our method outperforms all baselines on MultiHopQA and HotpotQA.

Method	F1	EM
DPR	62.3	48.1
ColBERT	65.7	51.4
Ours	71.2	56.8

Table 1: Main results on multi-hop QA benchmarks.

On MuSiQue, our method is competitive with the state of the art despite using a smaller retrieval model. Ablation studies show that the graph construction module contributes the most to performance gains, followed by the evidence aggregation strategy.

4. Conclusion

We demonstrated that reading order preservation is critical for two-column academic papers. Our results show consistent improvements when the correct reading order is maintained. Future work will explore more complex layouts.